

Sotirios Antonios Antonopoulos

Summary

As a Principal Data Engineer, my role centers on managing cross-functional teams and leading the design, development, and deployment of advanced data engineering and DataOps solutions. Working with a diverse team of engineers and data scientists, I bring expertise across multiple industries and have successfully built and optimized data pipelines, infrastructure, and analytics platforms for high-impact use cases. My role combines technical leadership with strategic insight, enabling me to serve as both a hands-on engineer and a trusted advisor to senior stakeholders, ensuring data solutions align with business objectives and deliver measurable value.

Work History

AI & Data Expert

02/2025 - Present

Accenture - European Public Prosecutor's Office (EPPO), Luxembourg

- Led the design and implementation of the AI validation framework for legal NLP systems (NER, document classification, multilingual OCR), ensuring compliance with EU AI Act principles.
- Built and evaluated Generative AI and RAG prototypes for legal-document intelligence and chatbot assistance across 22 EU jurisdictions.
- Developed data governance and validation pipelines—anonymization, benchmarking, and reproducible evaluation for multilingual legal corpora.
- Spearheaded the AI infrastructure strategy, including GPU cluster procurement (NVIDIA A100) and Nutanix-based on-prem environments for model training and inference.
- Collaborated with DPO and technical teams on responsible AI deployment, aligning transparency, accuracy, and privacy in production environments. Principal Data Engineer Element Mater

PRINCIPAL DATA ENGINEER

06/2024 - 08/11/2024

Element Materials - London, United Kingdom

- Served as part of the data leadership team on a greenfield data engineering initiative in a private equity environment, developing and deploying end-to-end data pipelines within a scalable Azure infrastructure.
- Designed an Azure-based Data Lake with Databricks and Terragrunt, streamlining data workflows and enhancing access for analytics and machine learning.
- Developed DataOps-driven ETL pipelines, automating data integration, processing, and validation to support robust, scalable document workflows.
- Leveraged Azure services on Azure Synapse and Azure Blob Storage to create scalable, cost-efficient data solutions, ensuring seamless integration and high performance across workflows.
- Integrated validation metrics across processing engines, providing critical insights for analytics and reporting.
- Aligned technical architecture with strategic goals through collaboration with senior stakeholders, ensuring optimal data strategy and value delivery.

LEAD DATA ENGINEER

08/2022 - 06/2024

SSnC - London, United Kingdom

- Technically Led multi-million dollar data engineering projects for major wealth management and investment clients, delivering scalable, high-impact data solutions that enhanced operational efficiency and drove substantial financial gains.
- Directed a cross-functional team of data engineers and DevOps specialists, aligning complex technical initiatives with business goals and fostering a culture of continuous improvement and agile collaboration.
- Architected and optimized end-to-end data pipelines across diverse sources such as Kafka, Postgres, and S3, implementing advanced DataOps practices with Apache Airflow to ensure seamless, real-time data flow, enhanced reliability, and robust scalability.
- Developed cost-effective ETL services using Spark, deployed in a private cloud on Kubernetes, significantly reducing operational costs and enabling efficient data processing for large-scale analytics.
- Strengthened data observability and governance by implementing Grafana for monitoring, metadata management, and integrating Superset with OKTA for secure and compliant data visualization across multiple business units.

- Established comprehensive quality assurance frameworks, ensuring data accuracy, integrity, and end-to-end traceability across the entire data lifecycle, from ingestion through to analytics and reporting.
- Led the modernization of financial reporting infrastructure, transitioning from legacy DB2 systems to a Delta Lake and Snowflake architecture, enhancing data accessibility, scalability, and enabling real-time analytics.
- Worked in a cross-cloud environment utilizing SSC private cloud, AWS (Lambda, S3, Glue, Athena, RDS) and Azure (Azure Data Factory, Azure Synapse, Azure Functions)

SENIOR DATAOPS ENGINEER 10/2021 - 08/2022 **DATAOPS ENGINEER** 03/2020 - 10/2021
IMG Arena - London, United Kingdom

- Developed, tested, and deployed ETL services focused on platform engineering, leveraging cloud environments for resource provisioning.
- Led key projects, such as Golf Strokes Gained analysis and automated content generation, and reported progress to the Head of Data and stakeholders, especially in AI/ML initiatives.
- Trained, deployed, and monitored supervised ML models in Production, with hands-on usage of MLflow for model lifecycle management.
- Actively mentored junior data engineers and contributed to a collaborative team culture through knowledge-sharing and supporting feature advancements in AI/ML.
- Engaged in architectural decisions, utilizing technologies like Apache Spark, AWS, Terraform, and TensorFlow, databricks

DATA SCIENTIST 09/2019 - 02/2020
Black Swan Data - London, United Kingdom

- Conducted Named Entity Recognition and entity disambiguation using BERT and Wikipedia taxonomies, while analyzing social media and Wikipedia lag to drive insights.
- Developed data engineering solutions and mentored team members on Databricks collaborating with stakeholders to align project outcomes.

DATA SCIENTIST 04/2018 - 09/2019
The Floow - Sheffield, United Kingdom

- Developed and productized a strategic crash detection algorithm for mobile applications, reducing accident reporting times by over 20% and enhancing risk assessment capabilities.
- Improved scoring models for assessing driving behavior, contributing to a 15% reduction in accident risk, and created automated data pipelines to streamline dataset generation for data scientists.

FREELANCER SOFTWARE ENGINEER 10/2016 - 04/2018
GroundZero Labs - Athens, Greece

- Part of scrum team developing an Omni-channel e-banking platform for Eurobank Ergasias, including an instant payments solution, enhancing digital banking services with a scalable, responsive architecture.
- Utilized front-end (ReactJS, Redux) and back-end (Spring Boot, MongoDB, MySQL, Redis) technologies to deliver a reliable, user-friendly platform.

FREELANCER DATA SCIENTIST 09/2015 - 10/2016
Mentat Innovations - Athens, Greece

- Developed streaming machine learning algorithms using Spark for POS providers, focusing on broken device detection and anti-fraud measures.
- Conducted data exploration and prototyped algorithms to optimize data processing, enhancing insights and functionality within the Mentat core software stack.

RESEARCH AND DEVELOPMENT ENGINEER 09/2014 - 09/2015
Nokia - Athens, Greece

- Developed software for the Virtual Gateway Platform (VGP) to support 4G telco cloud applications, implementing automatic rescaling of network elements using C/C++ and Python.
- Enhanced software reliability by applying Test Driven Development (TDD) in FCAPS and Operability applications.

Education

Master of Science, Artificial Intelligence University of Edinburgh - Edinburgh, United Kingdom	08/2014
Master of Engineering, Electrical and Computer Engineering Xanthi, Greece <i>Specialization in Digital Signal Processing and Electronics</i>	07/2012
Practical Generative AI Practical Generative AI Online - London, United Kingdom	11/2024